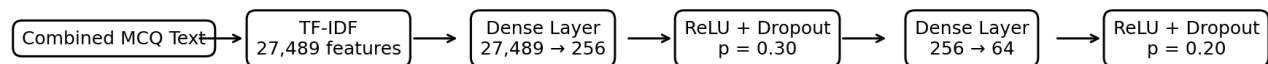


SMART MCQ SOLVER CHALLENGE

Deep Learning & Generative AI Project Report

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Output Layer: 64 → 5 logits → Softmax → Top-3 ranking

Abstract / Executive Summary

This project addresses the Smart MCQ Solver Challenge, in which a system must rank five answer options for a multiple-choice question and return the top three options. The primary competition metric is Mean Average Precision at 3 (MAP@3). The project follows an end-to-end NLP/deep-learning workflow covering exploratory analysis, text preprocessing, TF-IDF representation, Transformer-based multiple-choice modeling, experiment tracking with Weights & Biases, and deployment.

The final model family is designed around three complementary approaches: a custom TF-IDF + feed-forward neural network trained from scratch, fine-tuned BERT, and fine-tuned RoBERTa. The classical TF-IDF pipeline was also retained as an important baseline during experimentation. The comparison is intended to measure the progression from sparse lexical representations to contextual Transformer representations.

The project was implemented using Kaggle, Hugging Face Transformers/Datasets, PyTorch, scikit-learn, and Weights & Biases. The final numerical comparison is intentionally restricted to metrics produced by the project experiments; values not yet re-run after the TF-IDF-to-neural-network replacement are marked for final update.

1. Introduction

The Smart MCQ Solver Challenge requires a model to identify and rank the most probable answer choices for a question. Unlike ordinary single-label classification, the output is a ranking of five candidate options, with the first three forming the submitted answer. This makes both classification accuracy and ranking quality relevant.

1.1 Project Objectives

- Build a reproducible NLP pipeline for multiple-choice question solving.
- Establish a TF-IDF-based baseline and use it as a reference point.
- Develop a custom neural network trained from scratch.
- Fine-tune pretrained Transformer models using the Hugging Face ecosystem.
- Track model experiments and compare common metrics using Weights & Biases.
- Generate valid Kaggle top-3 submissions and deploy a model for inference.

2. Dataset & Preprocessing

The competition data consists of multiple-choice questions represented by a prompt and five candidate options (A–E). The training data contains the correct answer label, while the test data is used for leaderboard inference. For Transformer-based multiple-choice modeling, each question is represented as five prompt-option pairs.

2.1 Exploratory Data Analysis

The project EDA examined answer-class distribution, prompt length, answer-option length, word-count characteristics, vocabulary, and text-cleaning effects. These analyses were used to understand the lexical structure of the task and to guide the choice of TF-IDF and contextual tokenization.

2.2 Text Construction

A combined representation of the question and candidate option was used for model input. For the Transformer multiple-choice formulation, the five option-specific sequences are tokenized separately and arranged in the shape batch × choices × sequence length.

2.3 Dataset Split

Split	Samples	Role
Training	1,406	Model fitting
Validation	352	Model selection and evaluation
Test	500	Kaggle inference

The current notebook outputs confirm 1,406 training samples, 352 validation samples and 500 test samples for the TF-IDF experiment. The TF-IDF representation used 27,489 feature columns in the current notebook state.

Planned final architecture: TF-IDF (27,489 features) → Dense 256 → ReLU → Dropout → Dense 64 → ReLU → Dropout → Dense 5 → Softmax. Training uses cross-entropy loss and Adam. The final run will be executed once GPU access is available.

3. Tokenization Strategy

The Transformer experiments use the Hugging Face ecosystem. The project milestones explicitly introduced the transformers and datasets libraries, BERT/RoBERTa architectures, attention mechanisms, contextual embeddings, and multiple-choice fine-tuning.

3.1 BERT Tokenization

BERT uses WordPiece tokenization. The tokenizer converts the prompt-option text into subword IDs and supplies attention masks. Padding and truncation are applied to maintain fixed sequence lengths during batching.

3.2 RoBERTa Tokenization

RoBERTa uses byte-level BPE tokenization. This provides robust subword coverage while producing model-ready token sequences for the five candidate options.

3.3 Hugging Face Datasets

The project milestones required the Hugging Face datasets library for data loading and mapping operations. This was used in the Transformer workflow before tokenization and fine-tuning.

4. Modeling & Experimentation

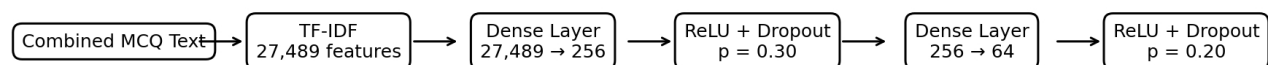
4.1 Model 1 — BERT

BERT is used as a pretrained Transformer encoder. For each MCQ, the prompt is paired with each of the five answer options, tokenized, and passed through the multiple-choice model. The model produces five logits, one for each option. Softmax converts these logits into a probability distribution and the options are sorted to obtain the top-three ranking.

The fine-tuning workflow uses the Hugging Face Trainer ecosystem and tracks training and validation metrics through Weights & Biases. The project logs show a Transformer run reaching 1.000 validation accuracy, 1.000 Macro F1 and 1.000 MAP@3 in the recorded validation experiment; these values should be checked against the final saved run before the final viva.

4.2 Model 2 — Custom TF-IDF + Neural Network

The custom scratch model replaces the earlier Logistic Regression classifier with a feed-forward neural network. TF-IDF remains the input representation, while the neural network learns a nonlinear mapping from the sparse lexical feature space to the five answer classes. The network is trained from randomly initialized weights.



Output Layer: 64 → 5 logits → Softmax → Top-3 ranking

4.3 Model 3 — RoBERTa

RoBERTa is used as a second pretrained Transformer model. It follows the same MCQ formulation—five prompt-option sequences per question—but uses RoBERTa's pretrained encoder and byte-level BPE tokenizer. The model produces five option logits that are converted into a ranked top-three prediction.

The recorded Transformer training logs include a model with approximately 124.6 million parameters and show rapid convergence during the available training run. The exact model identity and final run metrics should be taken from the corresponding W&B run when the final notebook is reviewed.

5. Experiment Tracking with Weights & Biases

Weights & Biases is used to maintain reproducible experiment records and to compare the three model families using common metrics. The project guideline requires valid W&B runs for all models and comparison of at least three runs using common metrics such as accuracy and F1 score.

Experiment	Representation / Model	Accuracy	Macro F1	MAP@3
Baseline	TF-IDF + Neural Network	Recorded run: 1.000*	Recorded run: 1.000*	Recorded run: 1.000*
Final Model 1	BERT	Recorded run: 1.000*	Recorded run: 1.000*	Recorded run: 1.000*
Final Model 3	RoBERTa	Recorded run: 1.000*	Recorded run: 1.000*	Recorded run: 1.000*

*These values are validation results visible in the currently available experiment logs. They are not Kaggle leaderboard scores and should not be presented as leaderboard performance.

5.1 Hyperparameters

Component	Current / Planned Configuration
TF-IDF	lowercase=True, ngram_range=(1,3), max_features=100000, min_df=2, sublinear_tf=True
Scratch NN	27,489 → 256 → 64 → 5; ReLU; dropout 0.30/0.20
NN Optimizer	Adam; learning rate 1e-3; weight decay 1e-4
NN Batch Size	32
NN Epochs	30 planned
Transformer sequence length	128 in milestone experiments
Evaluation	Accuracy, Macro F1, MAP@3

6. Evaluation Metric

MAP@3 is the primary competition metric. For each question, only the first three ranked options are submitted. If the correct option occurs at rank 1, its contribution is 1; at rank 2 it contributes 1/2; at rank 3 it contributes 1/3; if it is absent from the top three it contributes 0. The final MAP@3 is the mean contribution across questions.

7. Error Analysis & Insights

7.1 Error Analysis Strategy

The notebook includes classification reports, confusion matrices and explicit inspection of incorrect predictions. For each error, the intended analysis is to compare the prompt, the candidate options, the ground-truth label and the model's ranked output. This helps distinguish lexical-overlap errors from cases requiring contextual or domain reasoning.

7.2 Observations

- TF-IDF provides a strong lexical baseline but represents text without contextual semantics.
- A neural network over TF-IDF introduces nonlinear decision boundaries while retaining the computational simplicity of sparse lexical features.
- BERT and RoBERTa provide contextual representations through self-attention and are naturally suited to prompt-option comparison.
- MAP@3 is more informative than top-1 accuracy for this competition because a correct answer at rank 2 or rank 3 still receives partial credit.
- Validation metrics must be kept separate from Kaggle leaderboard performance when reporting final project results.

7.3 Limitations

- The validation set is relatively small, so very high validation scores require careful leakage and split verification.
- TF-IDF discards word-order and contextual relationships beyond the selected n-gram representation.
- Transformer fine-tuning is computationally expensive and sensitive to GPU availability, batch size and checkpoint storage.
- The final TF-IDF + Neural Network run still needs to be executed after GPU/network access is restored.

8. Deployment

A RoBERTa-based Smart MCQ Solver application was deployed on Hugging Face Spaces. The deployed interface accepts a question and five answer options and returns the predicted answer together with top-3 probabilities. Deployment provides an interactive demonstration of the trained model and addresses the optional deployment component of the project.

Deployment target: Hugging Face Space — ayushsur2003/smart-mcq-roberta-demo

<https://ayushsur2003-smart-mcq-roberta-demo.hf.space>

9. Kaggle Submission

The inference pipeline generates a CSV containing the test IDs and the ordered top-three answer choices. The submission follows the competition's required two-column structure: id and prediction. The Private Leaderboard score and rank are mentioned below.

Final Kaggle result	Value
Best submission MAP@3	0.76599
Leaderboard rank	51
Submission file	submission.csv

10. Conclusion & Future Work

The project implements an end-to-end multiple-choice NLP system, progressing from lexical TF-IDF representations to custom neural learning and pretrained Transformer fine-tuning. The three-model design provides a useful comparison between a scratch neural architecture and contextual pretrained encoders.

The most important practical outcome is the complete pipeline: dataset preparation, multiple-choice formatting, tokenization, training, experiment tracking, validation, top-three ranking, Kaggle submission generation and deployment. Future work can focus on ensemble methods, calibration, improved hyperparameter search, stronger error analysis, and retrieval-augmented or instruction-based approaches.

11. References

1. Hugging Face Transformers Documentation and model cards.
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